



STATISTICAL ANALYSIS OF ELECTORAL PATTERNS IN INDIAN DEMOCRACY (1977–2015)

STATISTICAL DATA ANALYSIS

SAMATRIX CONSULTING PVT LTD

GURUGRAM

contact@samatrix.io



Project Submission Guideline - Statistical Analysis of Electoral Patterns in Indian Democracy (1977–2015)

Learning Objectives:

1. **Understand Electoral Data Structures:**
 - Learn how Lok Sabha and Vidhan Sabha election datasets are structured.
 - Identify key columns such as votes, electors, party, constituency, and gender.
2. **Perform Data Cleaning and Transformation:**
 - Handle missing values using logic-based imputation.
 - Standardize party names and abbreviations.
3. **Compute Descriptive Statistics:**
 - Measure turnout rates, vote shares, number of candidates per seat, and gender distribution.
4. **Apply Statistical Techniques:**
 - Analyze vote margins, swings, and seat retention rates (incumbency).
 - Use hypothesis testing to compare ruling vs. opposition parties.
 - Calculate skewness and kurtosis of vote shares.
5. **Visualize Electoral Trends:**
 - Create histograms, boxplots, line plots, heatmaps, and Lorenz curves.
 - Interpret Gini coefficients and vote share correlations.
6. **Explore Regional and Temporal Trends:**
 - Identify top performing parties at national and state levels.
 - Analyze state-wise and year-wise changes in political dominance.
7. **Draw Policy-Level Insights:**
 - Reflect on political stability, fragmentation, and regionalization.
 - Evaluate the effectiveness of democratic participation across states and years.

Submission Type: Individual

What to Submit:

Each student must submit the following:

Jupyter Notebook (.ipynb file)

1. Filename: YourFullName_IPLStats.ipynb (e.g., AnanyaKumar_IPLStats.ipynb)
2. Your notebook must follow the steps and structure discussed in class. You are expected to complete all the analysis, visualizations, and statistical summaries as covered during the guided classroom sessions.
3. Provide detailed analysis of output of each step. Explain each plot and table in detail in 4-5 sentences
 - a. You may provide the detailed analysis in the same .ipynb file as markdown cell or separately in word/pdf format

Dataset:

Dataset for this analysis can be downloaded from the following link

<https://samatrix-data.s3.ap-south-1.amazonaws.com/Statistics-Project/ind-lok-sabha.csv>

<https://samatrix-data.s3.ap-south-1.amazonaws.com/Statistics-Project/ind-vidhan-sabha.csv>

Required Sections and Instructions:

1. Party Abbreviation Cleaning

- a. Create a mapping of each party name to its most frequently used abbreviation based on available data.
- b. Use the mapping to fill in missing party abbreviations in the dataset.
- c. Fill any remaining blanks with 'IND' to indicate independent candidates.
- d. Summarize what you observe—e.g., which parties appear most often and what that might indicate about election trends or major parties.

2. Standardize Abbreviations

- a. Replace Variations of Party Abbreviations to avoid treating the same party as different due to minor label differences.
- b. Update inconsistent abbreviations (e.g., change 'INC(I)' to 'INC') to ensure standard naming across the dataset.

3. Data Type & Missing Value Handling

- a. Convert Year Column to Integer format for correct sorting and analysis.
- b. Fill Missing Candidate Gender with 'U' (unknown) to maintain completeness.
- c. Drop Rows with Missing Total Vote Polled

4. Filling Missing Assembly Names

- a. Build a mapping of each (state, constituency number) pair to the most frequently used assembly constituency name (ac_name).
- b. Use this mapping to fill in missing ac_name values wherever possible.
- c. Report how many values are still missing after filling.
- d. Briefly explain how this improves data quality by using the most common name for each constituency.

5. Initial Data Exploration – Lok Sabha Dataset

- a. Check dataset size to see the number of rows and columns.
- b. Preview the data to understand its structure.
- c. Inspect data types and missing values for column types and for column-wise missing value counts.
- d. In your report, provide a short summary of what you observe—such as the number of records, key columns, and any data quality issues.

6. Handling Missing Values – Lok Sabha Dataset

- a. Fill missing candidate gender (cand_sex) with 'U' for unknown.
- b. Fill missing parliamentary constituency type (pc_type) with 'Unknown'.
- c. Check for any remaining missing values
- d. In your report, explain why these steps were taken.

7. Standardizing Party Abbreviations – Lok Sabha Dataset

- a. Replace inconsistent party abbreviation 'INC(I)' with the standard 'INC' to ensure uniformity.
- b. In your report, mention the importance of this step

8. Basic Summary Statistics – Vidhan Sabha Dataset

- a. Count the number of unique entries for key columns:
 - i. States (st_name)
 - ii. Years (year)
 - iii. Assembly constituencies (ac_no)
 - iv. Candidate names (cand_name)
 - v. Party abbreviations (partyabbre)
- b. In your report, include these counts and briefly mention what they tell you about the dataset's coverage and diversity.

9. Candidate Gender Distribution – Visualization

- a. Count how many candidates fall into each gender category using .value_counts().
- b. Display the gender distribution as a pie chart to show proportions of male, female, and unknown candidates.
- c. Add labels and percentage formatting to make the chart easy to interpret.
- d. In your report, include the pie chart and briefly describe which gender group is most represented in the dataset.

10. Trend of Average Candidates per Seat

- a. Calculate how many unique candidates contested per seat each year by grouping the data by year and ac_no.
- b. Compute the average number of candidates per seat for each year.
- c. Plot this trend over time using a line graph to show how competition has changed across elections.
- d. In your report, include the chart and describe whether the number of candidates per seat is increasing, decreasing, or stable over the years.

11. Voter Turnout Analysis by Year

- a. Calculate total votes polled and electors per constituency for each year using .groupby() and .agg(max).
- b. Sum these values at the national level for each year to get total turnout and electors.
- c. Compute voter turnout percentage using the formula:
- d. $\text{voter_turnout_percent} = (\text{total votes polled} \div \text{total electors}) \times 100$
- e. Plot a line chart showing how voter turnout has changed over the years.
- f. In your report, include the voter turnout chart and briefly describe whether turnout is increasing, decreasing, or fluctuating over time.

12. Top 10 Parties by Candidate Count

- a. Identify the top 10 parties with the highest number of candidates by using `.value_counts()` on the 'partyabbre' column.
- b. Create a horizontal bar chart to display these parties and their candidate counts.
- c. Use color coding (hue) for better visual distinction, but hide the legend to keep the chart clean.
- d. In your report, include the chart and mention any observations—e.g., which party fields the most candidates or if any regional parties rank in the top 10.

13. Identify Winning Candidates

- a. Select the winning candidate in each constituency by choosing the one with the highest `totvotpoll` (votes polled).
- b. Use `.groupby()` on year, state name, and constituency number, and extract the row with the maximum votes.
- c. Filter the winners by a specific state (e.g., Gujarat) to view regional results.
- d. In your report, show a sample of winners and explain how you identified them based on vote counts.

14. Compare Vote Share vs Seat Share – National Level

- a. Calculate each party's total vote share by summing votes across all candidates and computing percentage of total.
- b. Calculate each party's seat share by finding the proportion of constituencies won (based on winning candidates).
- c. Merge vote share and seat share into a single table.
- d. Select the top 10 parties by vote share for national-level comparison.
- e. Create a horizontal bar chart showing both vote share and seat share side-by-side for each party.
- f. In your report, include the chart and briefly explain whether top parties convert their vote share efficiently into seats. Provide your observations on High vote share vs High seat share

15. Gujarat – Top 3 Parties' Seat Trends Over Time

- a. Filter the dataset to include only winning candidates from Gujarat.
- b. Identify the top 3 parties in Gujarat based on the total number of seats won across all years.
- c. Limit the data to these top 3 parties for focused analysis.
- d. Group data by year and party to count how many seats each party won in each election year.
- e. Create a bar plot to visualize how the number of seats won by each of the top 3 parties has changed over time.
- f. In your report, include the chart and explain any patterns or shifts in political dominance across the years in Gujarat.

16. Vote Share Trend of Top 3 Parties in Gujarat

- a. Filter the dataset to include only data from Gujarat.

- b. Calculate total votes received by each party in each year (all candidates, not just winners).
- c. Compute total votes cast in Gujarat for each year.
- d. Calculate vote share percentage for each party per year using the formula:
 $\text{vote share} = (\text{party votes} / \text{total votes}) \times 100$
- e. $\text{vote share} = (\text{party votes} / \text{total votes}) \times 100$
- f. Identify the top 3 parties by total votes across all years in Gujarat.
- g. Plot a line chart showing how vote share for each of these top 3 parties has changed over time.
- h. In your report, include the chart and briefly describe how the political support for these parties has shifted over the years.

17. Vote Margin Analysis – Gujarat, Top 3 Parties)

- a. Calculate vote share percentage for each candidate using:
 $(\text{total votes polled} \div \text{total electors}) \times 100$.
- b. $(\text{total votes polled} \div \text{total electors}) \times 100$.
- c. Filter data for Gujarat to focus on a single state.
- d. Rank candidates within each seat based on total votes polled.
- e. Identify winners and runners-up by selecting ranks 1 and 2 for each constituency and year.
- f. Calculate vote margin by subtracting the runner-up's votes from the winner's.
- g. Find the top 3 winning parties in Gujarat by total number of wins.
- h. Filter vote margin data to include only these top 3 parties.
- i. Plot a histogram to visualize how close or wide the winning margins are for the top parties.
- j. Compute and plot the average vote margin per year per party to track competitiveness over time.
- k. Identify outlier years where the vote margin was unusually high or low using interquartile range (IQR).

I. Questions to Include in Report

i. Vote Margin Patterns

- ii. What does the distribution of vote margins tell us about the competitiveness of elections in Gujarat?
- iii. Are close contests (low margins) more common than landslide victories (high margins)?

iv. Party-wise Comparison

- v. Among the top 3 parties in Gujarat, which party tends to win by the largest average margin?
- vi. How do vote margins differ between these parties across different election years?

vii. Trend Over Time

- viii. Has the average vote margin increased, decreased, or remained stable over the years?
- ix. What might explain these trends—changing voter loyalty, stronger opposition, or shifting party dominance?

x. Outlier Analysis

- xi. Which years were identified as outliers in vote margin trends?
- xii. What political or social events might explain why the margins were unusually high or low in those years?

xiii. Competitiveness

- xiv. Based on vote margins and their changes over time, would you say Gujarat elections are becoming more or less competitive?

18. State-Level Voter Turnout and Constituency Analysis

- a. Calculate total votes polled and total electors for each constituency by grouping the data by state, year, and constituency number.
- b. Aggregate these constituency-level totals to the state level by summing total votes and electors for each state.
- c. Calculate voter turnout percentage for each state using:
- d. $(\text{total votes polled} \div \text{total electors}) \times 100$.
- e. Sort the states by voter turnout percentage to identify the highest and lowest participating states.
- f. Count how many unique constituencies were contested in each state using the number of unique constituency numbers (ac_no).
- g. Sort and display the states based on the number of constituencies to understand state-wise seat distribution.
- h. Plot two bar charts:
 - i. Voter turnout percentage by state
 - ii. Number of constituencies per state
- i. In your report, include both charts and explain which states show high or low participation, and reflect on what might contribute to these differences (e.g., population, geography, political activity).

19. Gender-Based Participation and Performance – Lok Sabha

- a. Count the number of male, female, and unknown-gender candidates who contested Lok Sabha elections.
- b. Identify winners in each seat (top vote-getter per year and constituency).
- c. Count how many winners belong to each gender category.
- d. Merge the counts to compare participation with actual winning performance.
- e. Calculate the win rate as a percentage for each gender:
- f. $(\text{Number of Winners} \div \text{Number of Candidates}) \times 100$
- g. Sort and display the results for comparison.
- h. Suggested Questions for Report (Gender-Based Electoral Analysis - Lok Sabha)
 - i. **Participation Patterns**
 1. Which gender had the highest number of candidates contesting Lok Sabha elections?
 2. Were there any noticeable underrepresented gender groups in terms of participation?
 - ii. **Winning Performance**
 1. Which gender secured the highest number of seats overall?
 2. Were there any gender groups with very few or no winners?
 - iii. **Win Rate Evaluation**
 1. Which gender had the highest win rate (success percentage)?
 2. Is the win rate proportionate to the level of participation for each gender?
 - iv. **Equity and Representation**

1. Does the data suggest any gender imbalance in electoral outcomes?
2. Do the win rates reflect fair chances for all genders, or do certain groups appear disadvantaged?

v. Critical Reflection

1. Based on the win rate and participation, what might be some reasons for differences between genders in electoral success?
2. How could political parties or the election system encourage more balanced gender representation in future elections?

20. Visual Analysis of Gender-Based Participation and Performance – Lok Sabha

a. Plot 1: Number of Candidates by Gender

- i. Create a bar chart showing how many candidates from each gender category (Male, Female, Unknown) contested Lok Sabha elections.
- ii. Use the column 'Number_of_Candidates' from your gender summary table.

b. Plot 2: Number of Winners by Gender

- i. Create a second bar chart showing how many winners belong to each gender group.
- ii. Use the 'Number_of_Winners' column to plot this.

c. Plot 3: Win Rate by Gender

- i. Create a third bar chart to display the win rate (in %) for each gender.
- ii. Use the 'Win_Rate (%)' column which you already computed

d. In your report:

- i. Include all 3 charts clearly labeled with titles and axes.
- ii. **Briefly describe the patterns observed:**
 1. Which gender has the highest participation?
 2. Which gender wins more often?
 3. Are there any surprising trends or imbalances?
- iii. Answer whether win rates are proportionate to participation, and what this could imply about representation or fairness in elections.

21. Seats Won by Top 5 Parties – Lok Sabha)

a. Step-by-Step Tasks:

- i. Identify winning candidates in each constituency and year by selecting the candidate with the highest number of votes (totvotpoll).
- ii. Count the number of seats won per party per election year using .groupby() and .size().
- iii. Find the top 5 parties based on total seats won across all years.
- iv. Filter the results to include only these top 5 parties for comparison.
- v. Create a line plot showing how many seats each top party won over time.

b. In your report:

- i. Include the line chart with proper labels and title.
- ii. Briefly describe:
 1. Which party has been consistently dominant?
 2. Are there any major changes or trends in party performance over time?

3. Does any party show a significant rise or decline in influence?
- iii. This analysis gives a clear picture of party-wise trends and helps explain shifting political landscapes

22. Vote Share Trend of Top 5 Parties – Lok Sabha

- a. Calculate total votes received by each party per year using `.groupby()` and `.sum()`.
- b. Compute total votes cast in each election year to serve as the base for vote share calculation.
- c. Merge the party-wise and year-wise totals to calculate vote share using the formula:
- d. $(\text{party votes} \div \text{total votes}) \times 100$
- e. Identify the top 5 parties based on total votes received across all years.
- f. Filter the data to include only these top 5 parties.
- g. Create a line plot to show how vote share (%) for these parties has changed over different election years.
- h. In your report:
 - i. Include the line chart with a proper title and axis labels.
 - ii. Briefly comment on:
 1. Which party has gained or lost vote share over time?
 2. Are there any crossover points or major shifts in political support?
 3. Does any party maintain a steady vote base?
 - iii. This helps you analyze national-level political trends through data.

23. Gini Coefficient of Vote Distribution – Lok Sabha)

- a. Define a function to calculate the Gini coefficient, which measures inequality in vote distribution among candidates.
- b. Calculate the Gini coefficient across all candidates, to understand how concentrated the vote shares are.
- c. Compute the Gini coefficient separately for each election year, using `.groupby()` and applying the Gini function to each year's vote totals.
- d. Plot the Gini coefficient over time to visualize whether vote inequality is increasing or decreasing.
- e. In your report
 - i. **Understanding the Gini Trend**
 1. What does the Gini coefficient tell us about vote distribution among candidates in Lok Sabha elections?
 2. How do high and low Gini values relate to political competition?
 - ii. **Historical Events & Gini Patterns**
 1. Why was the Gini coefficient lower in 1977? What political or social factors might have influenced this?
 2. What could explain the sharp rise in vote concentration between 1980 and 1996?
 - iii. **Sharp Changes in Vote Patterns**
 1. In 1998, there was a sudden drop in the Gini coefficient. What might have caused this shift toward more fragmented voting?
 2. How does this reflect the entry or rise of regional parties or independents?

iv. Bipolar Politics & Vote Consolidation

1. Between 1999 and 2014, the Gini coefficient steadily rose again. What does this suggest about the nature of electoral competition during this period?
2. How might the growth of national coalitions (e.g., NDA, UPA) have impacted vote concentration?

v. Long-Term Trends and Democratic Health

1. Based on the full trend from 1977 to 2019, do you think Indian Lok Sabha elections are becoming more or less competitive?
 2. What are the pros and cons of high vote concentration in a democracy?
- vi. Try to connect data trends with political history, electoral systems, and the broader implications of inequality in vote distribution.

24. Lorenz Curve of Vote Distribution – Lok Sabha

- a. Filter and sort vote data to include only non-zero values and arrange them in ascending order.
- b. Compute cumulative percentages for both:
 - i. Candidates (x-axis): Each candidate's position out of total.
 - ii. Votes (y-axis): Cumulative share of total votes received.
- c. Plot the Lorenz Curve, which visually shows the inequality in vote distribution.
 - i. The Line of Equality represents perfect equality (all candidates get the same votes).
 - ii. The Lorenz Curve shows the actual distribution—greater the curve bows below the line, the more unequal the vote distribution.
- d. Questions for Report
 - i. **Understanding the Shape of the Curve**
 1. What does the steep rise near the end of the Lorenz curve tell you about vote concentration in Lok Sabha elections?
 2. Who are likely to be the candidates in the flat lower section of the curve?
 - ii. **Measuring Inequality**
 1. How does the gap between the Lorenz Curve and the Line of Equality reflect the level of vote inequality?
 2. Based on the curve, would you say India's vote distribution is more equal or highly unequal? Why?
 - iii. **Electoral System Effects**
 1. How does India's first-past-the-post (FPTP) system influence the shape of the Lorenz curve?
 2. Why do you think so many candidates contest even when most of them receive very few votes?
 - iv. **Implications of Vote Concentration**
 1. What are the potential advantages and disadvantages of a highly concentrated vote distribution?
 2. Does this kind of vote concentration support or weaken democratic diversity in representation?

- v. Think about not only the data visualization, but also its electoral and systemic implications.

25. Skewness and Kurtosis of Vote Share – Lok Sabha

- a. Calculate vote share percent for each candidate using:
- b. $(\text{votes polled} \div \text{electors}) \times 100$, ensuring no division by zero.
- c. Remove any invalid entries (e.g., missing values or vote shares over 100%).
- d. Compute skewness to understand the symmetry of vote share distribution.
- e. Compute kurtosis to assess how peaked or flat the distribution is.
- f. Questions for Report
 - i. **Understanding Skewness**
 - 1. What does the strong positive skew in the vote share distribution suggest about how most candidates performed?
 - 2. Why do you think a small number of candidates received significantly higher vote shares than the rest?
 - ii. **Impact of Political Landscape**
 - 1. How does the presence of major political parties vs. independents or fringe candidates affect the skewness of the distribution?
 - 2. Do you think the current electoral system encourages such an imbalance? Why or why not?
 - iii. **Kurtosis Interpretation**
 - 1. The distribution is leptokurtic (kurtosis > 3). What does this tell you about vote share variation among candidates?
 - 2. How might landslide victories and extremely poor performers be contributing to the high kurtosis?
 - iv. **Implications for Electoral Representation**
 - 1. Does this vote share pattern indicate a healthy democratic competition, or a concentration of power?
 - 2. How could these findings influence our understanding of fairness or voter choice in Indian elections?
- v. Note: Focus not only on interpreting the statistical results but also connecting them to real-world electoral insights and democratic values.

26. Skewness & Kurtosis Over Time – Lok Sabha

- a. Group the data by election year, and for each year:
- b. Compute the skewness of vote share percentage to measure asymmetry.
- c. Compute the kurtosis of vote share percentage to assess the presence of extreme values or peaks.
- d. Store these statistics in a summary table with columns: year, Skewness, and Kurtosis.
- e. Questions for Report
 - i. **Historical Trends & Interpretation**
 - 1. Which election year showed the most balanced vote distribution? What political or historical context might explain this?
 - 2. Which year had the highest skewness and kurtosis? What does this say about electoral competition and dominance?
 - ii. **Systemic Effects**

1. How does India's first-past-the-post (FPTP) system influence vote distribution, based on the skewness observed across years?
 2. Do you think this system helps or hurts political diversity in elections? Use data trends to justify your answer.
- iii. Shifts in Electoral Dynamics
1. What major shifts in voting patterns can you detect from 1977 to 2014? Identify at least one period where vote inequality increased or decreased significantly and explain why.
 2. What does the dip in skew and kurtosis in 1998–1999 suggest about the nature of those elections?
- iv. Rise of Dominance or Diversity?
1. Do the more recent years (2009–2014) suggest a concentration of power in fewer parties or a return to balanced competition?
 2. Is high kurtosis (e.g., in 1991 or 1996) always a bad sign? Or can it reflect political clarity and strong mandates?
- v. Critical Reflection
1. What does the combination of skewness and kurtosis tell us about representation and voter voice in Indian democracy?
 2. Based on this analysis, would you say electoral contests are becoming more competitive or more one-sided over time?
- vi. Note: critically analyze statistical patterns while connecting them with India's political history and electoral system design.

27. Party-wise Skewness & Kurtosis of Vote Share – Lok Sabha

- a. Calculate total votes received by each party in each election year
 - i. Use `.groupby(['year', 'partyabbre'])` and `.sum()` to aggregate total votes per party per year.
- b. Compute total votes polled in each year
 - i. Aggregate overall votes cast per year using `.groupby('year')`.
- c. Calculate vote share (%) of each party in each year
 - i. Merge the above two datasets and compute
 - ii. $\text{vote_share_percent} = (\text{party_votes} / \text{total_votes}) \times 100$.
- d. Filter only those parties that have participated in at least 3 elections
 - i. This avoids unreliable statistics for parties that appeared only once or twice.
- e. Compute Skewness and Kurtosis for each party's vote share across years
 - i. Skewness shows whether a party had many small results and a few big spikes.
 - ii. Kurtosis shows whether vote share is stable or has extreme outliers.
- f. Present the summary table
 - i. Include columns: partyabbre, Skewness, Kurtosis, sorted by skewness.
 - ii. Print the top 5 most right-skewed parties (those with a few big results).
 - iii. Print the bottom 5 most left-skewed parties (those that may have declined over time).
- g. Questions for Report
 - i. **Right-Skewed Parties (One-time success stories)**

1. Which parties showed high positive skewness, and what does this suggest about their electoral success pattern?
 2. Why might a party gain high vote share in only one year, but perform poorly in others?
 3. What does it mean for a party to be a "one-hit wonder" in electoral politics?
 4. Are these parties mostly regional or national? How does that influence their ability to maintain vote share?
- ii. Left-Skewed Parties (Consistently strong with a setback)**
1. Which parties showed negative skewness, and what might have caused that one bad election?
 2. What does it say about a party's voter base and consistency if its skewness is negative but small in value?
 3. How does consistent performance over time help a party sustain relevance in Indian politics?
- iii. Other Questions**
1. Why is skewness a useful measure when analyzing party-wise vote share trends?
 2. What kind of political strategy might be risky for a party with high skewness or high kurtosis?
 3. Based on the skewness and kurtosis, which type of party (stable vs. spiky) would you prefer as a voter or as a political analyst? Why?

28. Party-wise Skewness & Kurtosis Visualization

- a. Select top and bottom 10 parties based on skewness:
 - i. Use the summary statistics table to pick the 10 most positively skewed and 10 most negatively skewed parties.
 - ii. Combine these into a single dataset for plotting.
- b. Create a bar plot for skewness:
 - i. Plot Skewness on the x-axis and Party on the y-axis.
 - ii. Use color shading to distinguish parties.
 - iii. Title your plot appropriately (e.g., "Party-wise Vote Share Skewness").
- c. Create a bar plot for kurtosis:
 - i. Plot Kurtosis on the x-axis and Party on the y-axis.
 - ii. Ensure this shows how sharply peaked or flat the vote share distribution is for each party.
- d. In your report:
 - i. Include both bar plots (skewness and kurtosis) as figures.
 - ii. Under each figure, write a brief interpretation, for example:
 1. "Parties with high skewness like SAD(M) had erratic performance with one big spike."
 2. "Parties with moderate kurtosis (~1–3) show some outliers, but not extreme fluctuations."
 - iii. Do skewness and kurtosis values align with the vote share trends you saw earlier?
 - iv. What does the visual pattern reveal that the table may not?

29. ANOVA Analysis of Party Vote Share Across States

- a. Choose one party (e.g., BJP, INC, AAP) to analyze.
- b. Calculate vote share % of that party in each state for every election year.
- c. Filter states with data for at least 3 elections.
- d. Run ANOVA test to check if the vote share varies significantly across states.
- e. Report:
 - i. F-statistic and p-value
 - ii. Conclusion: Is the difference significant?
- f. Include a boxplot of vote share by state.
- g. Questions for report
 - i. What does the ANOVA test help you determine in this context?
 1. → Explain in 1–2 lines how ANOVA checks whether vote share differs significantly across states.
 - ii. What is the meaning of the F-statistic and p-value in your result?
 1. → Interpret the numerical output of your ANOVA test.
 2. → Why is a low p-value (e.g., < 0.05) important?
 - iii. Based on your result, can you say that BJP's vote share is the same in every state? Why or why not?
 1. → Write a clear Yes/No with one-line justification.
 - iv. Which states appear to be BJP's strongholds and which are weak spots based on the boxplot?
 1. → Identify a few examples from the chart (e.g., high in Gujarat, low in Kerala).
 - v. Why is this analysis useful in real-world politics or strategy?
 1. → Write 1–2 lines on how this insight can be used (e.g., campaign targeting, voter analysis).

30. Statistical Comparison of Vote Share – Ruling vs Opposition Parties

- a. Brief Introduction (2–3 lines):
 - i. Explain the goal: Comparing vote shares of ruling vs opposition parties using statistical analysis
- b. Party Grouping Explanation:
 - i. List which party is treated as ruling (e.g., BJP).
 - ii. List opposition parties you included (e.g., INC, DMK, SP, etc.).
 - iii. Explain how you created the “group” column (Ruling or Opposition).
- c. Vote Share Calculation:
 - i. Confirm that you calculated vote share (%) for each party per year using the ECI method.
 - ii. Show the formula used: $(\text{party_votes} / \text{total_votes}) * 100$.
- d. T-Test Analysis:
 - i. Report the t-statistic and p-value.
 - ii. Clearly state if the result is statistically significant (p-value < 0.05).
 - iii. Write your interpretation:
 1. "Is the ruling party's vote share significantly higher than that of opposition parties?"
- e. Visualization:

- i. Include a boxplot comparing vote share distributions of the ruling vs opposition groups.
 - ii. Write 2–3 lines about what the plot shows visually
- f. Final Insight (2–3 lines):
 - i. Based on the result, what can you say about the relative strength of the ruling party vs opposition in terms of vote share?
 - ii. Why is this comparison useful in understanding Indian elections?

31. Party-Wise Vote Share Correlation Analysis (Lok Sabha Elections)

- a. Compute Total Votes per Party per Year
 - i. Group data by year and partyabbre.
 - ii. Sum the totvotpoll column.
 - iii. Rename result column to party_votes.
- b. Compute Total National Votes per Year
 - i. Group df_lok by year and sum totvotpoll.
 - ii. Rename result column to total_votes.
- c. Calculate Vote Share Percentage
 - i. Merge the two dataframes (party-wise and total votes).
 - ii. Compute vote share as $(\text{party_votes} / \text{total_votes}) * 100$.
 - iii. Create a new column vote_share_percent.
- d. Identify Top 5 Parties by Total Votes
 - i. Group by partyabbre and sum party_votes.
 - ii. Select the top 5 parties with highest total votes.
- e. Filter Dataset to Include Only Top 5 Parties
 - i. Filter the vote share dataframe using the top 5 party list.
- f. Reshape Data for Correlation Analysis
 - i. Pivot the dataframe so that:
 - 1. Rows = years
 - 2. Columns = party names
 - 3. Values = vote share percent
 - ii. Fill any missing values with 0.
- g. Calculate Correlation Matrix
 - i. Use .corr() on the pivoted dataframe to generate a correlation matrix between party vote shares.
- h. Visualize Using a Heatmap
 - i. Use seaborn.heatmap() to plot the correlation matrix.
 - ii. Use annotations, color maps, and labels for clarity.
- i. Analytical Questions for Report
 - i. Which two parties show the highest positive correlation in their vote share trends?
 - 1. What could be the reason for this? Do they operate in complementary regions or rise during the same political wave?
 - ii. Which two parties show the strongest negative correlation?
 - 1. Does this suggest one party's growth comes at the cost of the other? Why?
 - iii. Is there any pair of parties that show almost zero correlation?

1. What might explain the lack of relationship between their vote share trends?
- iv. How do independent candidates (IND) correlate with major national parties like BJP or INC?
 1. Do their vote shares increase or decrease together?
- v. What does the heatmap tell us about competition vs. coexistence?
 1. Are parties rising together, or competing in the same space?
- vi. How could a political strategist use this information?
 1. Give one practical insight from your correlation results for campaign or alliance planning.

32. Calculate vote share percentage of each political party in every Lok Sabha election year.

- a. Group by Party and Year
- b. Group by Year for National Vote Totals
- c. Merge the Two Datasets
- d. Calculate Vote Share Percent
 - i. Add a new column 'vote_share_percent'.
 - ii. Formula
 1. $\text{vote_share_percent} = (\text{party_votes} / \text{total_votes}) \times 100$
 2. This gives each party's vote share in that year as a percentage.
- e. Analytical Questions for the Report
 - i. Which parties had the highest vote shares in each year?
 1. Identify if a national party consistently dominates.
 - ii. Which years saw large vote share changes for any party
 1. Point out any sudden rise or fall for a party, and suggest historical/political reasons.
 - iii. Are there any years where multiple parties had similar vote shares?
 1. What does this suggest about competition and voter distribution?
 - iv. Can you spot any long-term trends?
 1. For example, is any party's vote share steadily increasing or decreasing?
 - v. What does the vote share tell us about the nature of Indian elections?
 1. Reflect on fragmentation, regional vs. national party performance, or the effect of coalitions.

33. Vote Share Trends of Top 5 Parties Over Time

- a. Identify the Top 5 Parties:
 - i. Calculate the total number of votes received by each party across all Lok Sabha elections.
 - ii. Pick the five parties with the highest total vote count.
- b. Filter the Data:
 - i. Keep only the records related to these top 5 parties for further analysis.
- c. Visualize Vote Share Trends:
 - i. Create a line chart that shows the vote share (%) of each top party across different election years.
 - ii. Make sure each party has a separate line with clear markers.

- d. Label the Chart Clearly:
 - i. Add appropriate title, axis labels, and legend for clarity.
 - ii. Ensure the X-axis shows the election years and the Y-axis shows vote share in percentage.
- e. Analytical Questions to Answer in Your Report
 - i. Which party had the most consistent vote share over the years?
 - ii. Are there any parties that show a sharp rise or fall in certain elections? What could be the reasons?
 - iii. Which party had the highest peak in vote share, and in which year?
 - iv. Did any party lose ground steadily over time?
 - v. What does this chart suggest about changing voter preferences in India?

34. Tracking Incumbency Retention in Lok Sabha Elections

- a. Rank Candidates in Each Constituency and Year:
 - i. Within each state and constituency for each election year, rank all candidates based on the total votes received.
 - ii. Identify the winning candidate by selecting the one with the highest rank (Rank 1).
- b. Track Winning Party Across Elections:
 - i. Sort the winning candidates chronologically by constituency.
 - ii. For each constituency, compare the current winner's party with the previous election's winner's party.
 - iii. Mark whether the seat was retained by the same party or not.
- c. Calculate Incumbency Retention Percentage:
 - i. For each election year, calculate the percentage of seats retained by the same party from the previous election.
 - ii. Convert this into a percentage format for easy interpretation.
- d. Visualize the Trend:
 - i. Plot a line graph showing the incumbency retention rate (%) over time (i.e., across different election years).
 - ii. Label the X-axis as "Year" and Y-axis as "Seats Retained (%)".
- e. Analytical Questions to Answer in Your Report
 - i. In which years did incumbents retain the highest proportion of seats?
 - ii. Are there election years where a major drop in retention occurred? What might have caused that?
 - iii. Does the trend suggest voter satisfaction with sitting governments or frequent desire for change?
 - iv. What political or historical events (e.g., major scandals, economic changes, political waves) could explain high or low retention years?
 - v. How does this help us understand the stability of party support in Indian constituencies over time?

35. Measuring Vote Share Swing for Top Political Parties

- a. Select the Top 5 Political Parties:
 - i. Use the top 5 parties already identified based on overall vote share across all election years.
 - ii. Filter the vote share dataset to include only these top parties.

- b. Calculate Vote Share Swing:
 - i. For each party, compute the year-over-year change (swing) in vote share.
 - ii. This is done by subtracting the previous election's vote share from the current year's vote share for that party.
- c. Create a Line Chart of Vote Share Swing:
 - i. Plot the swing values over the years for each party.
 - ii. The chart should clearly show whether a party's popularity increased or decreased in each election.
 - iii. Use a horizontal line at 0% to separate positive swings (gains) from negative swings (losses).
- d. Analytical Questions to Answer in Your Report
 - i. Which parties showed the largest positive swing in any year? What could have contributed to this rise?
 - ii. Identify the years where a party lost significant vote share. Are these linked to any political events or major leadership changes?
 - iii. Which party had the most volatile swing pattern (frequent ups and downs)?
 - iv. Are there any consistent gainers or losers across multiple elections?
 - v. What do these swings reveal about public sentiment or shifts in political alliances?